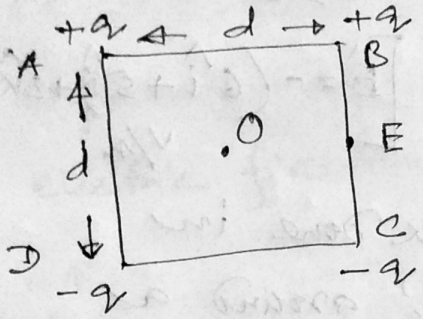
Home Work on Electric potential and potential difference :→

(1) 120 J of work has to be done against an existing electric field to take a charge of 0.01 C from 'A' to 'B'. Find $V_B - V_A$ = ? (Ans → 1200 volt)

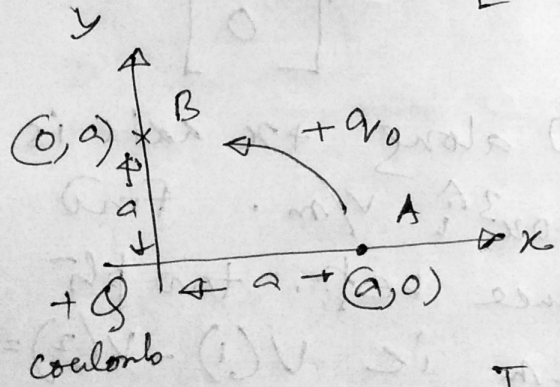
(2)



In figure Find work done to take a charge '+e' from centre 'O' to E (E mid point of side BC)

[Ans : 0 Joule]

(3)



Find work done by an external agent in transferring a charge '$+q_0$' from pt 'A' to pt 'B'.

[Ans Wext = 0]

(4) Electric potential at a point 'x' in a 1-dimensional electric field is $V = ax - bx^3$ [a, b = constants] Find 'x' where electric field strength 'E' is zero

[$x = \sqrt{\frac{a}{3b}}$]

(5) Potential at a pt (x,y) in an electric field is $V(x,y) = 3x + 2y^2$ volt. [x, y are in 'm']. Find magnitude of electric field at point $(3, 1)$

[Ans 5 N/C or 5 V/m]

(6) $V(x,y,z) = 6xy - y + 2yz$ volt.

find $E(\underset{m}{1}, \underset{m}{1}, \underset{m}{0})$. = ? $\left[\vec{E} = -(6\hat{i} + 5\hat{j} + 2\hat{k})\ V/m\right]$

(7) What is work done in revolving a charge 'q' around a fixed charge 'Q' in a complete circle of radius 'r'. [0]

(8) An electric field along $+x$ axis is given by $\vec{E} = 40x^3\hat{i}$ V/m. Find Potential difference betn. two pts $x = 1m$ & $x = 2m$ ie $V(1) - V(2) = ?$

[Ans 150 volts]

(9) A uniform electric field is $\vec{E} = 5(\hat{i} + \hat{j})$ V/m. Find work done by electric field in displacing +10C charge along y axis by 1m.

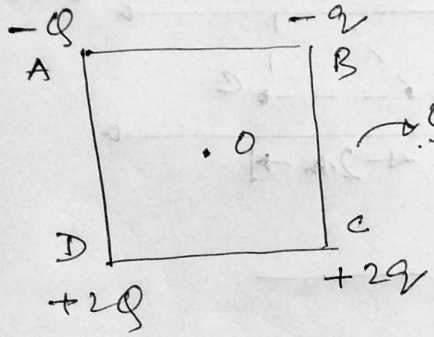
[Ans 50J]

(10) What is work done in revolving a charge 'q' around a fixed charge 'Q' in a complete circle of radius 'r'

[Ans → 0]

(11) In a uniform electric field a charge '+3C' experiences a force 3000 N. What is potential difference bet. two points separated by a distance '1cm' along electric field line. [10 volt]

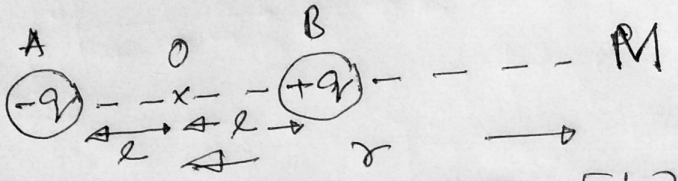
(12)



→ Square

If potential at centre O is zero then prove that 'Q = -q'.

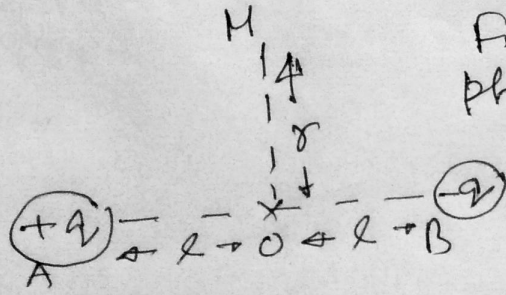
(13) (a)



OA = OB = l, OP = r, r >> l Find V_M = ?

{P = q × 2l = dipole moment of dipole} $\left[\frac{1}{4\pi\epsilon_0}\frac{P}{r^2}\right]$ (Ans)

(b)



Find potential at pt M, on ⊥ bisector of the dipole shown in fig?

[Ans 0]

(14) Give example in each case :

<u>Case 1</u> Electric field strength $\vec{E} = \vec{0}$ but potential $V \neq 0$

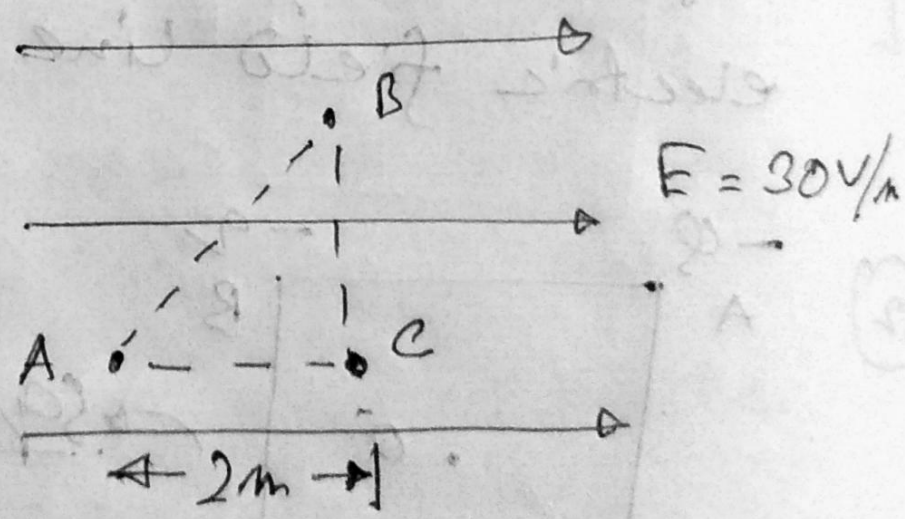
<u>Case 2</u> Electric field strength $\vec{E} \neq \vec{0}$ but potential $V = 0$.

(15) In constant electric field $E = 30 V/m$ as shown in fig Find

(i) $V_A - V_B$? (60V)

(ii) $V_A - V_C$? (60V)

(iii) $V_B - V_C$? (0)



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